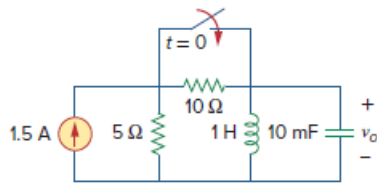


Problem

Find the output voltage $v_o(t)$ in the circuit of Fig. 16.64.

Figure 16.64:



Step-by-step solution

Step 1 of 6

Consider the switch is open for $t = 0^-$. Then, replace the inductor by short circuit and capacitor by the open circuit.

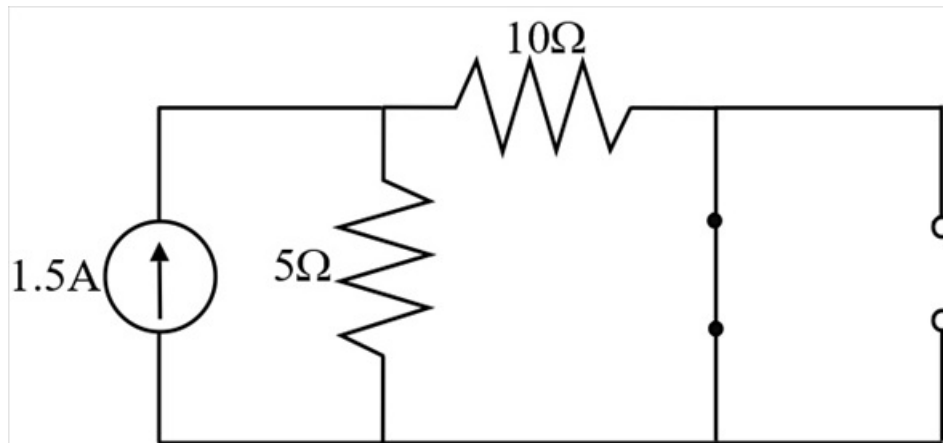


Figure 1

Step 2 of 6

Convert the current source in voltage source.

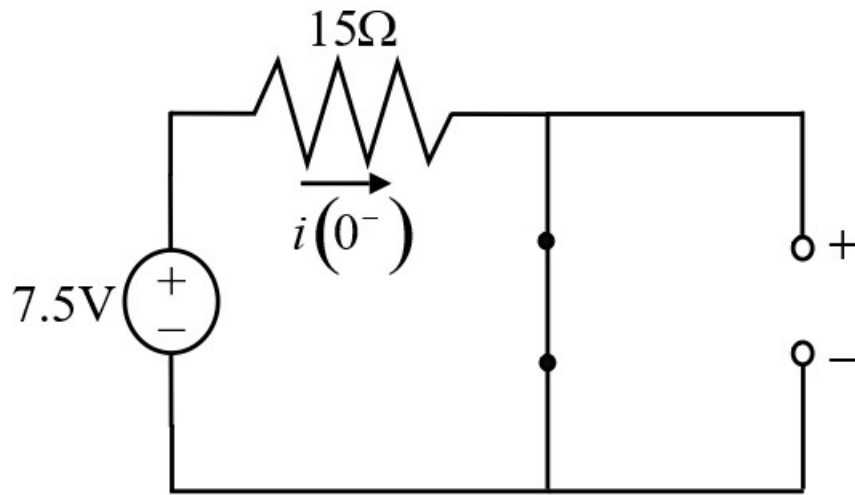


Figure 2

Step 3 of 6

Calculate the value of $i(0^-)$ and $v(0^-)$.

$$i(0^-) = \frac{7.5}{15}$$
$$= 0.5 \text{ A}$$

$$v(0^-) = 0 \text{ V}$$

Step 4 of 6

Consider the switch close at $t = 0$ and draw the s-domain equivalent of inductor and capacitor.

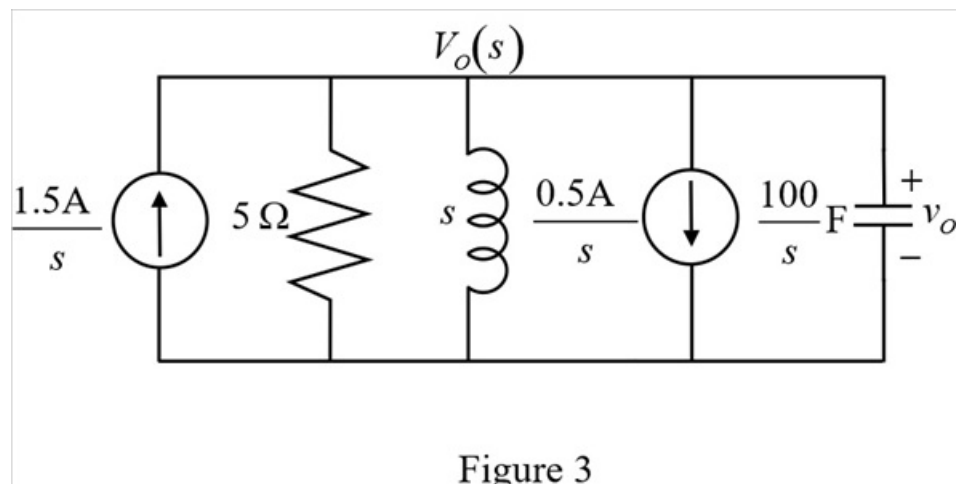


Figure 3

Step 5 of 6

Apply the nodal analysis in figure (3).

$$\begin{aligned}\frac{1.5 \text{ A}}{s} &= \frac{V_o(s)}{5} + \frac{V_o(s)}{s} + \frac{0.5 \text{ A}}{s} + \frac{s(V_o(s))}{100} \\ \frac{1}{s} &= \frac{20s(V_o(s)) + 100V_o(s) + s^2V_o(s)}{100s} \\ 1 &= \frac{20s(V_o(s)) + 100V_o(s) + s^2V_o(s)}{100}\end{aligned}$$

Rewrite the above equation.

$$\begin{aligned}V_o(s) &= \frac{100}{s^2 + 100 + 20s} \\ &= \frac{100}{(s + 10)^2}\end{aligned}$$

Step 6 of 6

Take the inverse Laplace transform of $V_o(s)$.

$$v_o(t) = [100te^{-10t}]u(t)$$

Hence the output voltage $v_o(t)$ is $\boxed{[100te^{-10t}]u(t)}$.